

CLWE-PRUNE: CROSS-LINGUAL WORST-GROUP EXCESS FOR MULTILINGUAL TEXT-TO-SPEECH COMPRESSION

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ABSTRACT

English-only layer selection can overlook degradation in other languages when compressing multilingual text-to-speech (TTS) models. We propose Cross-Lingual Worst-Group Excess Pruning (CLWE-Prune), which scores each candidate LLM block by the largest recognition-error excess caused by its removal from the dense model across English, Chinese, Japanese, and Korean, and prunes the lowest-scoring blocks. Recovery fine-tunes pruned models using generated speech-token sequences screened for text accuracy by offline automatic speech recognition (ASR). On CosyVoice3, the multilingual P_k configurations yield lower four-language average Whisper error than EN-WER P_k at all four matched pruning depths. With 400 verified sequences and 15 updates, recovery reduces Whisper Macro ER by 7.61–10.64% relative to the corresponding Base checkpoints at P_{12} , P_{15} , and P_{16} , and lowers SenseVoice Macro ER at all three depths. With 800 sequences and 40 updates, recovered P_{15} retains nine of 24 LLM blocks and has 26.03% fewer parameters than Dense-24. Its observed Whisper/SenseVoice Macro ER is 7.68%/8.73%, versus 7.86%/8.77% for Dense-24. A separate inference benchmark using a P_{15} checkpoint shows RTF decreasing from 0.633 for Dense-24 to 0.375 and first-packet latency decreasing from 2.600 to 1.572 s.

Index Terms— Multilingual TTS, Model Compression, Structural Pruning, Cross-Lingual Layer Selection

1. INTRODUCTION

Multilingual text-to-speech (TTS) systems increasingly use a shared Transformer backbone to synthesize speech across languages [1, 2]. Advances in neural TTS have improved acoustic modeling and waveform generation [3]; prompt-conditioned systems can synthesize speech for unseen speakers across languages [4, 5]; and speech-token models cast generation as autoregressive sequence prediction [6, 7]. This shared backbone, however, makes streaming inference costly: each autoregressive decoding step traverses the Transformer blocks in sequence. Removing entire blocks can shorten this path. Yet a block that appears dispensable for English may still be crucial for preserving the linguistic content of speech in languages such as Chinese, Japanese, and Korean. Aggregate error alone can obscure this risk: the dense model has different baseline

error rates across languages, while English is evaluated using word error rate (WER) and the other languages using character error rate (CER). We therefore focus on the *additional* error caused by pruning within each language, particularly the largest such increase across languages. Acoustic quality scores may remain stable even when linguistic content errors rise.

Structural pruning removes dimensions or layers [8, 9, 10]. SPADE applies WER-guided layer selection and distillation to LLM-based TTS [11], but its ranking is English-only and so cannot see regressions in Chinese, Japanese, or Korean. Averaging scores across languages is no safety net either: a large drop in one language can hide inside a flat mean, which is exactly why worst-group evaluation is preferred in robust model selection [12]. Pruning multilingual text models can also shift which language is harmed, as shown for text LLMs under mismatched calibration [13], and ASR-screened self-distillation has been used to reduce synthesis failures in neural-codec TTS [14]. To our knowledge, however, cross-lingual worst-group block selection combined with ASR-screened student-trajectory recovery has not been studied for pruned multilingual TTS.

Pruning also changes the student’s generation behavior. Standard distillation trains compact models on teacher information [15, 16], but sequences a dense teacher produces (or that come from recordings) do not match the paths a pruned student actually walks. On-policy distillation addresses this by learning from the student’s own generated sequences [17], and self-training work shows that generated supervision must be screened before use [18]. The risk here is concrete: a student trajectory that has already mispronounced a word, if used directly as training signal, teaches the pruned model to repeat that mistake. For TTS, an offline ASR recognizer can compare the decoded speech against the requested text and discard trajectories that fail this content check before they are used. This check runs only in training; it adds no cost to the deployed synthesizer.

We propose CLWE-Prune, which ranks layers by the worst-group excess error across four language groups and recovers each pruned model using ASR-screened student-generated speech-token trajectories, with the acoustic decoder held fixed. Our contributions are:

- We propose a cross-lingual layer-selection criterion based on the largest increase in recognition error relative to the dense model, improving multilingual content preservation over English-only ranking.

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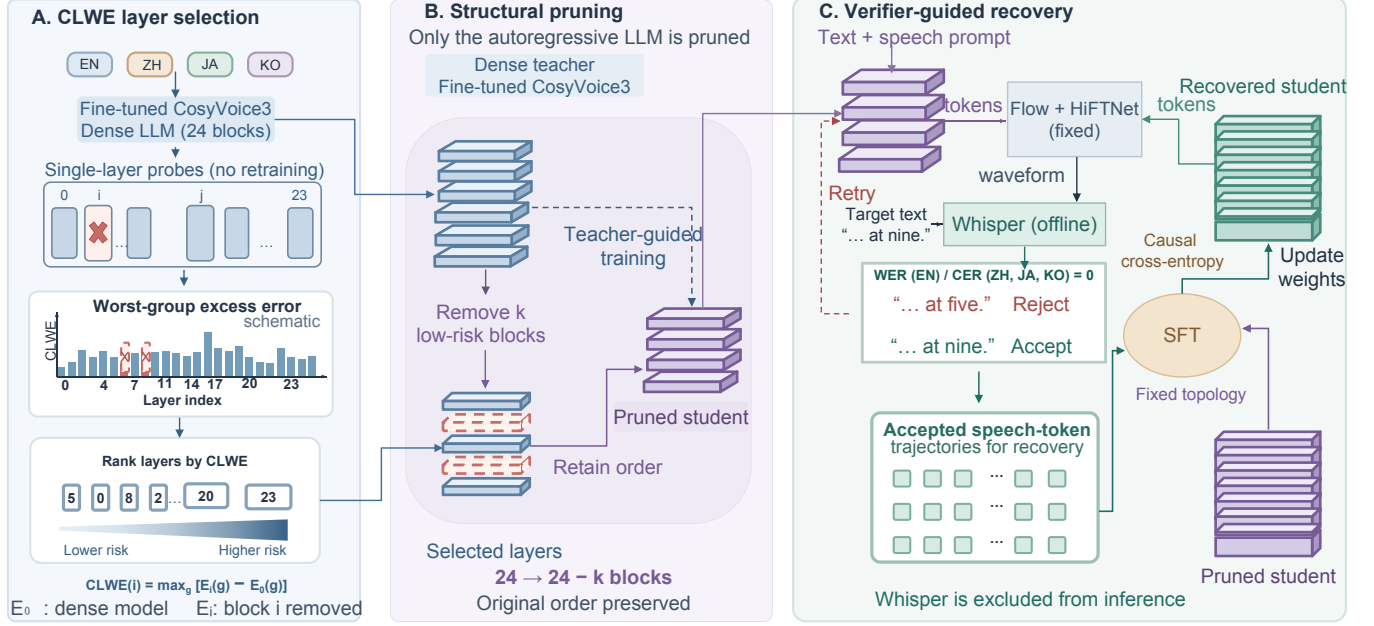


Fig. 1. Overview of CLWE-Prune. (A) Single-layer removal estimates cross-lingual worst-group excess risk. (B) Lowest-scoring blocks are removed and retrained to Base, with the acoustic decoder fixed. (C) The pruned student is recovered on ASR-screened speech-token trajectories, keeping the pruning topology fixed.

- We introduce verifier-guided recovery using ASR-screened, student-generated trajectories. This reduces recognition errors without changing the pruned architecture or acoustic decoder.
- Our recovered nine-layer model removes 26.03% of the dense model’s parameters while achieving comparable recognition accuracy and a $1.69\times$ streaming real-time-factor speedup.

2. METHOD

Figure 1 summarizes three stages: single-layer risk estimation, physical pruning, and verifier-guided recovery. We rank CosyVoice3’s 24 autoregressive LLM blocks by their cross-lingual worst-excess error and remove the lowest-risk blocks. The retained blocks preserve their original order and inherit their weights from Dense-24. Under the matched-selection protocol, the pruned LLM undergoes continuation training guided by a frozen Dense-24 teacher; the checkpoint with the lowest validation loss becomes the Base model. Recovery then fine-tunes this Base model on ASR-screened, student-generated speech-token trajectories. The teacher used to construct Base is distinct from the Dense-Teacher recovery control, which supplies recovery targets. The flow-matching acoustic model and HiFTNet vocoder remain unchanged throughout.

2.1. CLWE-Based Layer Selection

As shown in Fig. 1(A), we estimate single-block removal risk on a multilingual layer-selection dataset. We group samples by language, with one group per language. Thus,

$$\mathcal{G} = \{g_{\text{EN}}, g_{\text{ZH}}, g_{\text{JA}}, g_{\text{KO}}\}.$$

To probe block i , we remove only that block from Dense-24 and evaluate the remaining model without retraining. All single-block probes use the same inference and scoring procedure. Let $E_i(g)$

and $E_0(g)$ denote the recognition errors of the probe and Dense-24, respectively, on language group g .

Following SPADE’s WER-based layer-importance approach [11], the English-only baseline ranks block i by $R_i^{\text{EN}} = E_i(g_{\text{EN}})$. Subtracting the constant $E_0(g_{\text{EN}})$ from every English score would leave this ranking unchanged. This baseline compares layer-selection criteria; it does not reproduce the full SPADE training pipeline.

CLWE instead measures the additional error caused by removing each block within each language:

$$\begin{aligned} \Delta_i(g) &= E_i(g) - E_0(g), \\ R_i^{\text{CLWE}} &= \max_{g \in \mathcal{G}} \Delta_i(g). \end{aligned} \quad (1)$$

Subtracting $E_0(g)$ accounts for each language’s Dense-24 baseline, while the maximum makes the largest observed error increase determine the block’s risk. This subtraction does not make English WER statistically equivalent to the other languages’ CER. A lower R_i^{CLWE} indicates a safer block to remove.

We rank blocks in ascending order of risk, breaking ties by their original indices. For a pruning budget of k , we physically remove the first k blocks in this ranking, as shown in Fig. 1(B). The ranking is computed once and reused across pruning depths, so the removal sets are nested and the retained blocks keep their original order. Single-block probes do not capture interactions among jointly removed blocks; consequently, their ranking is a selection heuristic, not a bound on the final pruned model’s error.

2.2. Verifier-Guided Student-Trajectory Recovery

As shown in Fig. 1(C), each pruned Base model generates speech-token trajectories $\mathbf{z} = (z_1, \dots, z_T)$ conditioned on text x and prompt c . Generation uses temperature 1.0, $\text{top-p}=0.8$, and $\text{top-k}=25$. The fixed flow-matching acoustic model and HiFTNet vocoder decode the tokens into speech. After language-specific text

normalization, Whisper evaluates content using WER for English and CER for Chinese, Japanese, and Korean.

Zero-error candidates are prioritized. When an input has no acceptable candidate, we generate additional candidates or draw from a shared reserve text pool. Among candidates passing the content check, speaker similarity, UTMOS, and duration ratio guide selection; these quality measures do not replace the ASR criterion. Candidate generation and screening are performed offline.

Each pruned student Base first generates 400 speech-token trajectories, from which we select recovery targets using the verifier. We fine-tune the student on these targets and report its step-15 checkpoint as SFT400/15. This checkpoint then generates a second set of 400 verifier-screened targets.

For SFT800/40, we reinitialize the student from its original pruned Base checkpoint and fine-tune it on both sets of targets. We report the resulting step-40 checkpoint. Thus, SFT400/15 generates the second target set but is not the initialization for SFT800/40. In the P12 Base-generated control, both sets of targets are generated by the original P12 Base, while the final model is trained from that same Base.

Let $\mathcal{D}_{\text{rec}}$ contain the accepted trajectories. For sequence j in a minibatch $\mathcal{B}$, set $\tilde{z}_{j,t} = z_{j,t}$ for $t \leq T_j$ and $\tilde{z}_{j,T_j+1} = \text{EOS}$. With $h_{j,t} = (x_j, c_j, \tilde{z}_{j,<t})$, we optimize

$$\mathcal{L}_{\text{rec}} = -\frac{1}{N_{\mathcal{B}}} \sum_{j,t} \log p_{\theta_P}(\tilde{z}_{j,t} | h_{j,t}). \quad (2)$$

Here $\sum_{j,t}$ ranges over $j \in \mathcal{B}$ and $t = 1, \dots, T_j + 1$, while $N_{\mathcal{B}} = \sum_{j \in \mathcal{B}} (T_j + 1)$ counts valid targets, excluding masked text and padding positions. The parameters θ_P belong to the pruned LLM. Recovery uses a learning rate of 3×10^{-6} and mixed precision. Teacher forcing trains the LLM to predict the accepted sequences; Whisper provides no training gradients.

2.3. Experimental Setup

Data. The calibration set uses 400 utterances per language–corpus group from LibriTTS (English) [19], AISHELL-3 (Chinese) [20], and Emilia (Japanese and Korean) [21]. The `strict1200` test set has 300 utterances per language. Models share test texts and prompts but generate separate waveforms. Recovery uses 100 or 200 targets per language for SFT400/15 or SFT800/40.

Models and training configurations. Dense-24 is the unpruned 24-block model. Removing k LLM blocks leaves $24 - k$ blocks. Multilingual configurations selected by CLWE at this depth are labeled P_k , and the English-only pruning baseline is labeled EN-WER P_k . Table 1 compares both at $k \in \{6, 8, 10, 12\}$. Table 2 extends the multilingual sweep to P_{13} – P_{17} . Its separate inference benchmark reports Dense-24, P_{12} , P_{15} , and P_{16} .

Table 3 compares recovery from P_{12} , P_{15} , and P_{16} Base checkpoints, which are separate from the Table 2 sweep. Verifier-SFT is evaluated at both 400/15 and 800/40 for all three depths. At P_{12} , additional controls include Base-Generated-SFT800/40, Selected-Original400/15 using original-recording tokens, and Random-Candidate400/15 using unverified student targets with separately assigned same-language prompts. Dense-Teacher-SFT400/15 uses verified dense-model targets at all three depths. Dense-24 Teacher is the unpruned reference.

Evaluation metrics. Content accuracy is evaluated using Whisper large-v3 [22] and SenseVoiceSmall [23]. WER is used for English, while CER is used for Chinese, Japanese, and Korean. We report macro-averaged error rate (Macro ER), computed as the mean

of the four language-level error rates. Table 3 averages decoding seeds 1986–1988 per fixed checkpoint. For a model evaluated alongside Dense-24, the test worst-group excess is $\max_g [E_{\text{model}}(g) - E_{\text{Dense}}(g)]$.

We report UTMOS predicted quality [24] and mel-cepstral distortion (MCD) in the recovery comparison. The high-compression sweep also reports CAM++ speaker embedding cosine similarity (SECS) [25]. MCD uses 24-kHz WORLD 24-order mel cepstra (excluding c_0) aligned with fastDTW. Higher UTMOS/SECS and lower MCD are preferred. UTMOS also ranks candidates; these automatic scores do not establish perceived naturalness. A separate architecture benchmark measures streaming real-time factor (RTF), first-packet latency, and peak allocated CUDA memory.

Table 1. English-WER versus multilingual CLWE layer selection. Bold marks values where P_k outperforms EN-WER P_k at the same depth.

Model	Macro↓	EN/ZH↓	JA/KO↓	UTMOS↑/MCD↓
EN-WER P6	7.92	2.16/12.17	12.12/5.26	3.47/8.32
P6	7.67	2.75/11.67	11.92/4.34	3.48/8.28
EN-WER P8	7.91	2.23/12.67	11.93/4.81	3.47/8.32
P8	7.65	2.04/12.01	12.16/4.37	3.48/8.26
EN-WER P10	8.47	2.81/11.97	13.85/5.27	3.48/8.27
P10	7.78	2.53/12.28	11.45/4.88	3.47/8.29
EN-WER P12	8.97	2.64/12.10	16.25/4.90	3.49/8.31
P12	8.09	2.42/12.38	12.51/5.04	3.48/8.32

2.4. Results and Analysis

Table 2. High-compression quality sweep and separate architecture inference benchmark. L denotes retained blocks. Macro ER is Whisper Macro ER.

Model	Layers	Macro ER↓	UTMOS↑	SECS↑	MCD↓
P13	11	8.36	3.48	0.80	8.31
P14	10	8.65	3.48	0.80	8.36
P15	9	8.75	3.48	0.80	8.35
P16	8	11.31	3.47	0.80	8.40
P17	7	13.17	3.48	0.80	8.46

Model	Params (M)↓	First pkt. RTF↓	First pkt. (s)↓	Peak CUDA (MiB)↓
Dense-24	859.19	0.633	2.600	3382
P12	680.24	0.434	1.779	2697
P15	635.50	0.375	1.572	2530
P16	620.59	0.377	1.535	2468

Table 1 compares layer rankings at matched depths. The multilingual P_k models lower Whisper Macro ER by 0.25, 0.26, 0.69, and 0.88 points at P6, P8, P10, and P12. The largest reported language error also falls by 0.25, 0.51, 1.57, and 3.74 points, respectively. This descriptive maximum mixes English WER with the other languages’ CER. P12 Japanese CER (16.25% to 12.51%) shows the non-English regression missed by English-only ranking. P10 Chinese CER rises by 0.31 points, so gains are not uniform. Mean-Excess and Max-Raw checkpoints were not evaluated, so this comparison does not isolate the maximum operation or Dense-baseline subtraction. The matched comparison ends at P12.

Table 3. Recovery performance on `strict1200`. W and SV denote Whisper large-v3 and SenseVoiceSmall. Bold marks the best value within each pruning depth; Dense-24 Teacher is the unpruned reference.

Recovery Model	W-Macro↓	SV-Macro↓	EN W/SV	ZH W/SV	JA W/SV	KO W/SV	UTMOS↑	MCD↓
P12 (Base)	8.15	9.23	2.64/4.11	12.36/10.86	12.36/14.19	5.23/7.76	3.4884	8.3143
P12 + Verifier-SFT400/15	7.53	8.40	2.61/3.75	11.79/10.56	11.16/12.41	4.58/6.87	3.5241	8.2449
P12 + Verifier-SFT800/40	7.24	8.32	2.36/3.68	11.51/10.45	10.76/12.39	4.35/6.74	3.5496	8.2541
P12 + Base-Generated-SFT800/40	7.62	8.33	2.55/3.88	12.01/10.39	11.51/12.40	4.39/6.64	3.5411	8.2609
P12 + Selected-Original400/15	8.28	9.34	2.59/4.12	12.25/10.71	12.50/14.50	5.77/8.05	3.4746	8.2967
P12 + Random-Candidate400/15	7.62	8.53	2.54/3.98	12.20/10.67	11.33/12.59	4.41/6.89	3.5133	8.2388
P12 + Dense-Teacher-SFT400/15	7.68	8.57	2.61/4.14	7.16/5.32	15.53/16.83	5.41/7.99	3.5404	8.3762
P15 (Base)	8.91	9.73	3.00/4.40	12.17/10.88	13.54/15.07	6.94/8.56	3.4788	8.3662
P15 + Verifier-SFT400/15	8.00	8.59	2.78/4.02	12.27/10.78	12.39/12.48	4.54/7.08	3.5280	8.2871
P15 + Verifier-SFT800/40	7.68	8.73	2.72/4.24	12.09/10.81	11.09/12.44	4.83/7.43	3.5543	8.2912
P15 + Dense-Teacher-SFT400/15	8.82	9.44	2.93/4.55	7.74/5.81	18.72/18.65	5.90/8.76	3.5282	8.4419
P16 (Base)	9.87	10.80	3.65/5.30	12.69/11.09	15.58/17.13	7.57/9.69	3.4721	8.4053
P16 + Verifier-SFT400/15	8.82	9.50	3.38/4.54	12.35/11.02	14.02/14.07	5.54/8.35	3.5345	8.3495
P16 + Verifier-SFT800/40	8.32	9.27	3.21/4.78	12.36/11.12	12.09/13.20	5.62/7.96	3.5688	8.3476
P16 + Dense-Teacher-SFT400/15	10.16	10.80	3.78/5.28	8.01/6.16	21.75/22.44	7.08/9.31	3.5249	8.4836
Dense-24 Teacher	7.86	8.77	2.13/3.39	12.21/10.67	13.72/14.48	3.39/6.52	3.4724	8.3113

Table 2 reports available multilingual P_k configurations, not a controlled depth-only ablation. Whisper Macro ER rises from 8.36% at P13 to 8.75% at P15 and 11.31% at P16, while UTMOS and SECS barely change. This motivates testing recovery at P_{15} and P_{16} . In the separate architecture benchmark, RTF falls from 0.633 for Dense-24 to 0.375 for P_{15} (1.69 $\times$). First-packet latency falls from 2.600 to 1.572 s. These efficiency checkpoints are distinct from the quality and recovery checkpoints. Peak allocated CUDA memory falls from 3382 to 2530 MiB. The P_{16} architecture has a slightly lower first-packet latency (1.535 s) but no further RTF gain (0.377). The measured quality and efficiency trends favor P_{15} among these configurations.

Relative to their respective Base models in Table 3, Verifier-SFT400/15 reduces Whisper/SenseVoice Macro ER by 0.62/0.83, 0.91/1.14, and 1.05/1.30 points at P12, P15, and P16, respectively. UTMOS increases and MCD decreases at each depth without changing parameter counts. SenseVoice does not select targets, so its improvement supports a content gain beyond the Whisper screening metric. Among 24 per-language ASR values (three depths, four languages, two recognizers), 23 decrease. The sole exception is P15 Chinese Whisper CER (12.17% to 12.27%). This breadth shows that the macro gains are not driven by one language. UTMOS also ranks accepted targets.

SFT800/40 further improves both ASR measures at P12 and P16. At P15, Whisper improves from 8.00 to 7.68 but SenseVoice rises from 8.59 to 8.73. Relative to their own Base checkpoints, the P15 and P16 SFT800/40 models reduce Whisper/SenseVoice Macro ER by 1.23/1.00 and 1.55/1.53 points, respectively. At P15, UTMOS increases from 3.4788 to 3.5543 and MCD decreases from 8.3662 to 8.2912. These automatic acoustic scores move with the recognition gains, although UTMOS is also used to rank accepted recovery candidates. The recovered P_{15} model has 635.50M total parameters, versus 859.19M for Dense-24, and observed Macro ER of 7.68/8.73 versus 7.86/8.77. Its worst-group Whisper excess over Dense-24 is still 1.44 points in Korean (4.83 versus 3.39). English WER is also 0.59 points higher. Thus, the lower macro average does not imply uniform language-level preservation.

At P12, Selected-Original is slightly worse than Base (8.28/9.34

versus 8.15/9.23), while Random-Candidate reaches 7.62/8.53 and verified student recovery reaches 7.53/8.40. Student-generated targets give the larger observed gain, consistent with the on-policy rationale [17]. Different prompts prevent assigning the smaller verified-random gap solely to ASR screening. With 800 targets, mixed Base/auxiliary generation reaches 7.24/8.32 versus 7.62/8.33 when Base generates all targets, with separate training runs. Dense-Teacher at P15 lowers Chinese Whisper CER from 12.17% to 7.74% but raises Japanese CER from 13.54% to 18.72%. At the same 400-target/15-update setting, verified student targets yield lower Whisper/SenseVoice Macro ER than Dense-Teacher targets by 0.15/0.17, 0.82/0.85, and 1.34/1.30 points at P12, P15, and P16. The Japanese Whisper CER gap is especially large at P15 (12.39% versus 18.72%) and P16 (14.02% versus 21.75%), whereas the teacher targets improve Chinese. Thus the lower macro errors do not merely reflect a single favorable language.

3. CONCLUSION

We presented CLWE-Prune, which ranks LLM blocks by worst-group recognition-error excess across four languages and recovers pruned models using ASR-screened student trajectories, with the acoustic decoder held fixed. At all four matched depths, the multilingual P_k configurations have lower Whisper Macro ER than EN-WER P_k ; at P_{12} , Japanese CER is 12.51% versus 16.25% for EN-WER P_{12} . With Verifier-SFT400/15, 23 of 24 per-language ASR values improve relative to the corresponding Base checkpoints across P_{12} , P_{15} , and P_{16} . Recovered P_{15} with SFT800/40 has 26.03% fewer parameters than Dense-24 and slightly lower observed Whisper/SenseVoice Macro ER. Relative to the corresponding Base checkpoints, dense-teacher targets reduce Chinese errors but increase Japanese errors at all three depths; under the matched 400/15 budget, verified student targets yield lower macro errors. Language-level gaps remain, notably in English and Korean. Joint selection and recovery, broader language coverage, and human evaluation are left for future work.

4. COMPLIANCE WITH ETHICAL STANDARDS

This study used existing publicly available speech corpora and involved no new data collection or interaction with human participants. No additional ethical approval was required.

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